Posterior Integration and Volatility Inputs in WTI Asian Options

Integración posterior y volatilidad en opciones asiáticas WTI

Abstract

Objective: Compare the effect of posterior integration with that of choosing a different volatility input when pricing WTI average-price options.

Methods: We use a second-order expansion, 10 million simulated return histories, and two separate comparisons with option settlement prices.

Results: Across 310 October-2026-expiry observations, posterior integration and posterior-mean plug-in prices differ by only $0.000745/bbl on average. For 2,381 observations across 15 dates, historical-volatility mean absolute error (MAE) is $2.6187/bbl, versus $0.1075/bbl with earlier option-implied volatility. A separate 97,341-observation comparison also favors earlier implied volatility (MAE $0.2184 versus $0.8040/bbl). Bayesian updating of implied volatility does not reliably improve on carrying forward its last value.

Recommendation: Compare volatility inputs before adding model complexity.

Limitations: Results depend on settlement proxies, model assumptions, few original valuation dates, and same-date inputs in the larger comparison.

Originality: We distinguish the effect of posterior integration on option prices from the effect of volatility inputs and uncertainty about pricing errors.

Conclusion: Changing the volatility input affects pricing errors more than posterior integration in these tests.

Keywords: Bayesian option pricing; Asian options; WTI crude oil; volatility uncertainty; implied volatility.

JEL classification: C11; C15; C58; G13; G17; Q40.

Resumen

Objetivo: Comparar el efecto de la integración posterior y de la fuente de volatilidad al valuar opciones asiáticas sobre WTI.

Métodos: Empleamos una expansión de segundo orden, 10 millones de series simuladas y dos comparaciones con precios de liquidación.

Resultados: En 310 observaciones del vencimiento de octubre de 2026, la integración posterior y el uso de la media posterior de volatilidad difieren $0.000745 por barril en promedio. En 2,381 observaciones de 15 fechas, el error absoluto medio (MAE) es $2.6187 por barril con volatilidad histórica y $0.1075 con volatilidad implícita anterior. Otra comparación de 97,341 observaciones favorece la volatilidad implícita anterior (MAE de $0.2184 frente a $0.8040 por barril). Un modelo bayesiano no supera claramente el uso de la última volatilidad implícita.

Recomendación: Comparar fuentes de volatilidad antes de añadir complejidad.

Limitaciones: Los resultados dependen de precios de referencia, supuestos de valuación, pocas fechas originales y datos contemporáneos en la comparación ampliada.

Originalidad: Distinguimos integración posterior, elección de volatilidad e incertidumbre sobre los errores.

Conclusión: La fuente de volatilidad afecta más los errores que la integración posterior.

Palabras clave: valuación bayesiana de opciones; opciones asiáticas; petróleo WTI; incertidumbre de volatilidad; volatilidad implícita.

Clasificación JEL: C11; C15; C58; G13; G17; Q40.

1. Introduction

What matters more when pricing an option on the average price of crude oil: allowing for uncertainty about volatility, or choosing a volatility estimate informed by option prices? WTI Average Price Options (APOs) hedge exposure to an arithmetic average of crude-oil futures settlements, rather than a single price at expiry. Their value depends on both the pricing model and its volatility input. Bayesian methods account for uncertainty about that input, but greater computational complexity does not necessarily produce a more accurate conditional price.

We separate three questions. First, how much does averaging option prices over the volatility posterior change the price itself? We call this posterior integration (PI), and compare it with pricing at the posterior mean (PM). Second, how much does the chosen volatility input affect pricing errors? Third, how accurately can we describe errors relative to observed settlement prices? The PI–PM difference, the spread of model prices across the posterior, and an interval for settlement-price errors are different quantities; none should stand in for another.

We study these questions using 10 million simulated return histories and two separate analyses of market data. Each analysis has its own pricing model and information cutoff. The first uses individual Barchart APO histories, a CL futures-curve-based one-factor Asian option approximation, 310 October-expiry contract-date observations in August–September 2026, 2,381 date-ordered conditional repricing observations across 15 dates, and a 253-observation cross-instrument check using earlier LO vanilla-WTI implied volatilities. The second uses Databento AAO settlement records and a *different*, simplified pricing model that treats the terminal arithmetic average as lognormal. It compares prices with 97,341 observations classified by the provider as actual settlements across 189 dates. It also examines Bayesian estimates based on earlier implied volatility and intervals for subsequent settlement-price errors.

The results point to a practical distinction. In the original study, integrating over historical volatility uncertainty changes the point price very little. Using earlier option-implied volatility instead of the tested historical estimates yields much smaller errors relative to settlement-price proxies. The separate, longer AAO study finds a similar ranking of volatility inputs, but Bayesian updating of earlier implied volatility does not reliably

outperform carrying forward the most recent observation. Its pricing-error intervals also fall below their stated coverage levels in parts of 2026. These results are conditional: some settlement prices may reflect exchange valuation models, and the longer study uses other same-date option strikes to estimate its pricing inputs. The two studies use different models and must not be pooled as one sample.

The contribution is an empirical comparison of what changes a WTI average-price option valuation: integrating volatility uncertainty or changing the volatility input. We also distinguish uncertainty in model prices from uncertainty in observed pricing errors. The comparison provides a practical test of whether additional Bayesian modeling improves conditional pricing. We keep the limitations of settlement data and model assumptions explicit.

2. Literature review

Asian options depend on a price average rather than a single terminal price. Geometric-average approaches and conditioning approximations are long-established tools for computing arithmetic-average option values (Curran, 1994; Kemna & Vorst, 1990). REMEF has published related Asian-option work: Matías González et al. (2019) study a stochastic optimal consumption/investment framework with Asian-option valuation, whereas Ortiz Ramírez & Jiménez Preciado (2026) apply realized volatility and mean reversion to the CEMEXCPO setting. Neither compares the effect of averaging over a volatility posterior with the effect of using different volatility inputs to price WTI average-price options.

Bayesian option pricing is also well established. Guidolin & Timmermann (2003) study learning and option-price dynamics, while Rombouts & Stentoft (2014) report a small effect of parameter uncertainty on pricing errors in their equity-option setting. More recently, Sagor et al. (2026) propagate Bayesian uncertainty into illustrative European-option prices, including a WTI example. Those studies do not make the same comparison for WTI average-price options. Our contribution is not a new method of posterior integration. It is a direct comparison of posterior integration, earlier option-implied volatility, and a simple benchmark that carries forward the last observed implied volatility.

WTI APOs provide a different test setting. Market-specific models may account for stochastic convenience yields, jumps or richer futures dynamics (Ewald et al., 2023; Schwartz, 1997; Shiraya & Takahashi, 2011). We hold the one-factor option-pricing benchmark fixed in the original analysis, so the influence of volatility information is not conflated with changes in the dynamics of the commodity futures curve. This comparison focuses on the volatility input rather than on selecting a more complex model for crude-oil futures.

The distinction between historical and option-implied information is important. Gilder & Tsiaras (2020) study forecasts of *subsequent realized crude-oil volatility* extracted from option prices. We study errors in pricing WTI *average-price options* at later valuation dates. Volatility inferred from option prices need not match volatility estimated from historical futures returns, particularly when the two estimates describe different underlying quantities. Consequently, a large historical-versus-implied pricing difference by itself neither identifies a variance risk premium nor demonstrates that Bayesian inference fails.

Those studies examine different pricing and volatility-forecasting questions. Here we hold the pricing model fixed and compare the effect of posterior integration with the choice of volatility input. We also distinguish uncertainty in model prices from errors relative to observed settlement prices.

3. Methodology

3.1 Physical-measure inference and risk-neutral valuation

Historical log returns are modeled under the physical measure as

$$R_i \sim \mathcal{N}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2 \Delta t\right],$$

with baseline priors

$$\mu \sim \mathcal{N}(0,1), \qquad \sigma \sim \text{InvGamma}(2,0.1), \qquad \sigma > 0.$$

We draw from the posterior distribution of $(\mu, \log\sigma)$ using random-walk Metropolis. In the large synthetic experiment, we instead integrate out μ analytically given σ. The resulting one-dimensional calculation avoids Monte Carlo sampling error from MCMC in that experiment.

Pricing is carried out separately under the risk-neutral measure. For futures maturity u, the maintained benchmark is

$$dF_t^u = \sigma F_t^u \, dW_t^Q,$$

where u indexes futures maturity and the same Brownian motion W^Q drives all maturities. Under this model, futures returns at different maturities have perfect instantaneous correlation. Physical drift does not enter risk-neutral valuation. Holding these dynamics fixed allows direct comparisons of posterior integration and volatility inputs.

For an arithmetic-average option with fixing dates $t_1, \ldots, t_M$,

$$A_T = \frac{1}{M}\sum_{j=1}^{M} F_{t_j}^{(1)},$$

and the conditional call value is

$$C_t^Q(\sigma) = B(t,T)\,\mathbb{E}^Q[(A_T - K)^+ \mid \mathcal{F}_t; \sigma],$$

where $B(t,T)$ is the discount factor, K the strike, and $\mathcal{F}_t$ the valuation-date information set. Put values are defined analogously. In the arithmetic-average definition above, $F_{t_j}^{(1)}$ is the first-nearby WTI futures settlement on fixing date t_j under the applicable exchange contract rules (CME Group, 2021). At a valuation date within the averaging period, earlier fixings are known. We simulate only the remaining fixings, using the corresponding contracts in the observed CL futures curve as initial values.

We price the baseline historical-volatility model by Monte Carlo. For the larger date-ordered comparisons using alternative constant volatilities, we use Curran's conditioning approximation (Curran, 1994). We check numerical accuracy with independent Monte Carlo and scrambled Sobol calculations.

3.2 Posterior integration and price uncertainty

Let $\mathcal{D}$ denote the historical return data, $C^Q(\sigma)$ the conditional risk-neutral option value, and $\bar{\sigma} = \mathbb{E}[\sigma \mid \mathcal{D}]$. We compare posterior-integrated (PI) and posterior-mean plug-in (PM) prices:

$$C_{PI} = \mathbb{E}[C^Q(\sigma) \mid \mathcal{D}], \qquad C_{PM} = C^Q(\bar{\sigma}).$$

A second-order expansion around $\bar{\sigma}$ gives

$$C_{PI} - C_{PM} \approx \frac{1}{2} C^{Q\prime\prime}(\bar{\sigma}) \mathrm{Var}(\sigma \mid \mathcal{D}).$$

The formula shows how posterior variance and the curvature of the pricing function determine the approximate PI–PM difference. Its sign depends on the curvature, and the approximation is most reliable when higher-order terms are small over the posterior range.

The posterior distribution for volatility also gives a distribution of model prices,

$$C^Q(\sigma), \qquad \sigma \sim p(\sigma \mid \mathcal{D}).$$

Its dispersion is approximately

$$\mathrm{SD}(C^Q(\sigma) \mid \mathcal{D}) \approx |C^{Q\prime}(\bar{\sigma})| \sqrt{\mathrm{Var}(\sigma \mid \mathcal{D})}.$$

The two quantities respond differently to volatility uncertainty. The PI–PM difference is approximately proportional to the posterior variance of volatility, whereas the standard deviation of model prices is approximately proportional to the posterior standard deviation of volatility. Neither quantity describes errors relative to future market prices or accounts for a misspecified pricing model.

3.3 Synthetic experiment

The synthetic experiment varies the posterior variance of volatility and the curvature of the pricing function in the second-order approximation above. It contains 10 million simulated return histories, generated with root seed 2468 and distributed over 35 combinations of historical sample size (21 to 1,260 observations) and annual volatility (0.10 to 0.80). We evaluate 168 option states varying maturity, moneyness, and the fraction of the arithmetic average already fixed. For each history and option state, we calculate the PI–PM difference, posterior volatility variance, local price curvature, and the second-order approximation. The volatility posterior is evaluated on a 4,097-point grid; computations are batched without changing the model or pricing grid.

This design shows when posterior integration has a larger effect and checks the second-order approximation separately from errors against market prices.

3.4 WTI Average Price Options and market data

We begin with Barchart price histories for individual WTI Average Price Option contracts. The archive covers 213 contracts and 11,179 contract-date observations, with expiries from September 2026 to June 2029. We compare model prices with the Barchart end-of-day `Latest` field, treated as a settlement-price proxy. It has not been independently certified as an official final exchange settlement or executable transaction quote. We include available calls and puts and calculate moneyness separately for each valuation date.

For every observation, we identify the fixings already known and the futures contracts used for the remaining fixings. We initialize those contracts with the futures prices observed on the valuation date. This matters because the first-nearby contract can change during the averaging month.

We estimate historical volatility from WTI futures returns available by the valuation date. The baseline uses the continuous/front-month `CL=F` return series. A robustness check reconstructs a first-nearby return history contract by contract from CL settlements and excludes returns spanning contract switches.

3.5 Date-ordered conditional pricing comparisons

We compare volatility inputs by repricing APOs on later dates. The option observations used to estimate implied volatility are dated before each valuation date in the archived files. However, we cannot independently verify when the vendor first published the records or whether they were subsequently revised. Each price uses the futures curve, known fixings, and discount factor on the valuation date, while implied volatility comes from earlier option observations. Historical-volatility estimates use the specified return-data cutoff. This is a date-ordered conditional pricing comparison, not proof that all records were available in real time. The sample contains 2,381 contract-date observations across seven expiries and 15 valuation dates.

The baseline integrates conditional prices over a volatility posterior estimated from an expanding return sample available at the stated valuation cutoff. The exercise evaluates conditional pricing errors using that day's observed futures curve and fixings, not prices forecast before those inputs became available. Historical alternatives use rolling windows of 63, 126, or 252 returns, an exponentially weighted moving average (EWMA) with a 63-business-day half-life, and a horizon-matched Gaussian GARCH(1,1) forecast. For GARCH, parameters are estimated from earlier returns; a valuation-date futures return updates conditional variance, which is then forecast over the remaining fixings. The GARCH comparison therefore also uses end-of-day information from the valuation date. It is not a forecast using only earlier-day information. We express these forecasts as the constant annualized volatility that gives the same variance for the remaining average under the one-factor pricing model. Full comparisons are in Online Supplement Section S1 and Table S1.

We obtain contract-level APO implied volatilities by inverting the Curran pricing approximation. For each expiry, the previous-day implied-volatility smile is estimated on the latest available earlier date. Another specification estimates a smile from all earlier

dates, with greater weight on recent observations. Both depend on moneyness and option type. Neither uses the target-date APO settlement to estimate volatility.

To assess the value of smile fitting, we also carry forward each contract's most recent implied volatility. We reprice with the valuation-date curve, discount factor, and fixing schedule. This same-contract benchmark covers 2,366 observations.

Pricing performance is summarized by mean error, mean absolute error (MAE), and root mean squared error (RMSE). We use a cluster bootstrap that resamples valuation dates, keeping observations from the same date together.

3.6 Cross-instrument validation

The first comparison estimates implied volatility using earlier APO observations. We therefore add a cross-instrument check using standard monthly American WTI futures options (CME product LO) from Databento. We match LO option settlements to the underlying CL futures and invert prices using a Cox–Ross–Rubinstein futures-option tree.

For each APO target date, only LO observations from earlier dates are used. Implied-volatility smiles are fitted separately for the underlying CL futures contracts needed by the October-2026 APO fixing schedule. The resulting maturity-specific volatilities are mapped into the maintained scalar-volatility APO pricing model. We compare 253 APO contract-date observations over 10 target dates for which both external specifications can be evaluated within the moneyness range observed in earlier LO data. Online Supplement Section S4 and Table S4 report checks of the constant-volatility mapping and the number of steps in the Cox–Ross–Rubinstein tree.

This check asks whether the lower pricing errors also hold when implied volatility comes from a different WTI option product. It does not measure a variance risk premium.

3.7 Separate Databento pricing comparison

We examine the volatility-input comparison in a second dataset of AAO settlement records from Databento. This is *not* an extension of the earlier Curran model using CL fixing paths. The datasets, pricing methods, and information available for valuation differ. The Databento AAO definitions and settlement statistics cover January 2020–September 2026; admissible evaluation dates run from January 2023 through September 2026. We select every fifth available source date and apply the same contract eligibility rules throughout, obtaining 189 valuation dates, 1,689 expiry-date groups, and 111,585 contract-date observations. Of these, 97,341 are classified by the data provider as actual settlements and 14,244 as theoretical. All headline error comparisons below use the 97,341 observations classified as actual.

We estimate the risk-neutral expected terminal average and an effective discount factor by fitting put–call parity to a separate set of matched call and put strikes on the valuation date. Those strikes are excluded from the options used to evaluate pricing errors. We then apply the Black formula to a *lognormal terminal arithmetic average*. This is a reduced-form approximation, not the Curran method applied to a path of first-nearby CL fixings. Because the pricing inputs use other AAO settlement prices from the same date, this is a conditional comparison, *not* a forecast made entirely with earlier information.

The historical-volatility input is estimated from earlier changes in the expected AAO average inferred from put–call parity for the same expiry. It is therefore neither a history of CL futures returns nor a risk-neutral volatility estimate. Posterior integration (PI) and posterior-mean plug-in (PM) use the same return-proxy posterior. Rolling and exponentially weighted proxies are evaluated on exactly the same option targets. The simple implied-volatility benchmark carries forward volatility inferred from near-the-money AAO option prices on an earlier date, using the *same* terminal-average pricing model. No implied volatility from the valuation date is used as an input to this benchmark. All six methods are evaluated on the same contracts and dates.

A second exercise uses only *strictly earlier* near-the-money AAO implied volatilities to model changes in log implied volatility for each expiry. Bayesian normal-inverse-gamma updating yields a Student-t posterior predictive log-volatility distribution: `bayes_IV_PI` averages option prices over 256 predictive volatility draws; `bayes_IV_PM` prices at their mean. Both are compared with the price obtained by carrying forward the last observed implied volatility for the same options. The chronological comparison is reported separately for 2023–2024 development dates and 2025–September 2026; the latter dates had already been inspected during initial work and are not an untouched confirmatory holdout for every subsequent diagnostic.

In a separate analysis, we construct 90% and 95% prediction intervals for errors between observed settlement prices and model prices, using either a Bayesian normal-inverse-gamma model fitted to past errors with equal weight across dates or rolling empirical quantiles of past errors. It uses a 63-date maximum window, requires at least 15 earlier dates per put/call-side and maturity stratum, and evaluates the intervals using observed coverage, width, and the interval score. These are *not* posterior credible intervals for an Asian payoff, nor posterior price bands from the volatility parameter in the earlier October-expiry study. We resample by valuation date or calendar month for paired comparisons, rather than treating contracts on the same date as independent observations.

3.8 Computational implementation

The computations are implemented in Python using NumPy, SciPy and pandas for inference, simulation and market-data processing, and Matplotlib for figures. A public source-only repository contains the pricing library, experiment scripts, a synthetic demonstration, numerical tests, and source-provenance records. The original vendor observations and internal market-dependent outputs are retained privately. The synthetic calculation integrates physical drift analytically conditional on volatility. For empirical validation, estimated volatility inputs follow the documented time cutoffs, and errors are resampled by valuation date to retain within-date dependence.

3.9 AI-assisted manuscript preparation

OpenAI ChatGPT assisted with research software development, literature organization, language editing, and drafting and revision of the manuscript. The author remains responsible for the methods, numerical calculations, independent checking and interpretation of the results.

4. Results and discussion

4.1 When does posterior integration matter?

Figure 1 shows when posterior integration changes option prices most in the simulated settings. The largest differences occur with short estimation windows, high volatility, long maturities, few realized fixings, and pronounced price curvature. Across the 10 million synthetic histories, the maximum observed absolute difference is 0.1565 for a normalized $100 forward, and the highest 99th percentile among the simulated scenario groups is 0.1273. These extremes exceed the differences observed in the October-2026 sample. The maximum is sensitive to the number of simulations, while the 99th percentile is less affected by rare outcomes.

The second-order approximation tracks the simulated differences closely: the correlation between cell-level mean PI–PM differences and the curvature–variance term is 0.9995. Regressions constrained to pass through the origin also have slopes near one in the most extreme settings. These results support posterior variance and price curvature as the principal local determinants of the difference.

The PI–PM difference generally declines as the share of realized fixings increases. The change is not always steady: in Panel C, the 252-day case initially increases because known fixings affect both the remaining exposure to volatility and the curvature of the price function. The difference nevertheless tends toward zero as the averaging period ends.

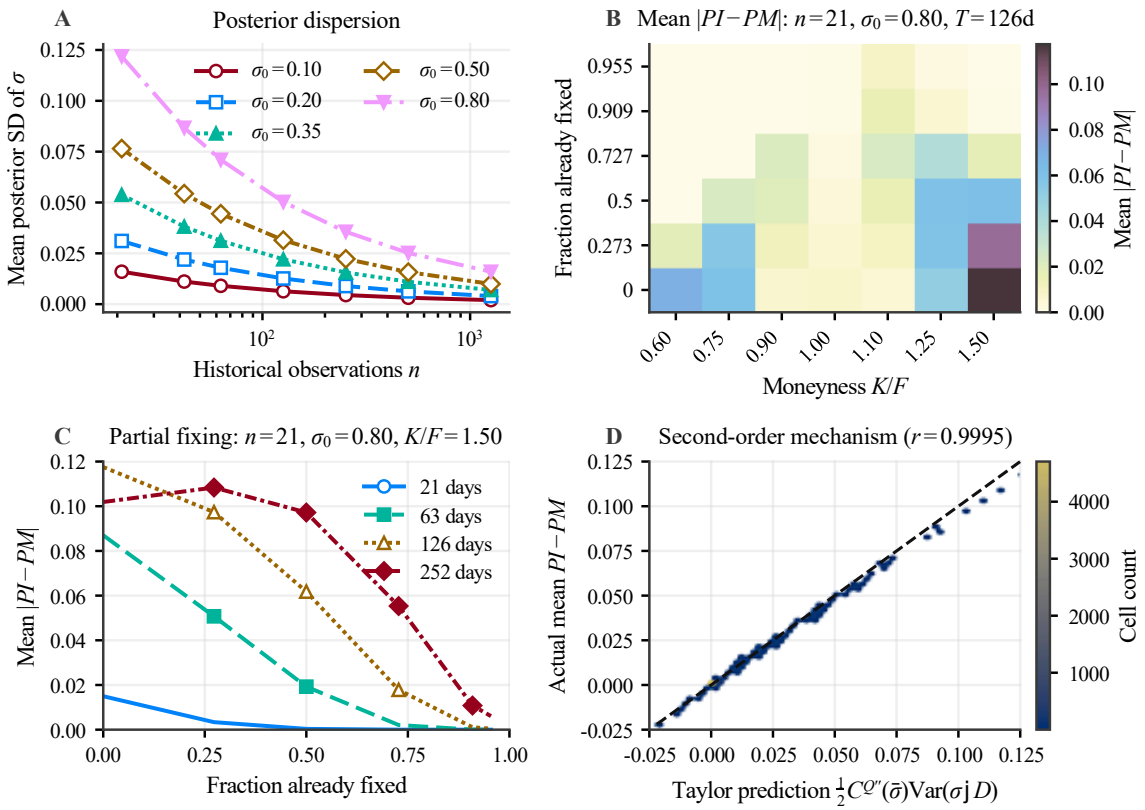


Figure 1. Posterior integration across 10 million synthetic return histories and WTI-like option states (root seed 2468). Panel A shows posterior dispersion by historical sample size. Panels B and C show the absolute PI–PM difference across moneyness and the realized-fixing fraction. Panel D compares mean PI–PM differences for 5,880 scenario cells with the second-order approximation described in the methodology; the cell-level correlation is 0.9995. The color bar shows the number of scenario cells per hexagonal bin.

Source: Authors' calculations from the recorded experiment outputs.

4.2 Posterior integration in the WTI sample

The sample of options expiring in October 2026 contains 310 contract-date observations across 12 valuation dates from August 25 to September 10, 2026. Table 1 shows nearly identical pricing errors for PI and PM (RMSE 0.5177 and 0.5183, respectively). Their mean absolute price difference is only 0.000745 dollars per barrel.

The central 95% posterior interval for conditional model prices has mean width 0.2733 dollars per barrel, approximately 367 times the mean absolute PI–PM difference. The model prices vary across posterior volatility draws, even though averaging those prices barely changes the point estimate. This posterior interval does not describe forecast errors against observed settlements or the value of an executable trade. Both methods underprice the reported October settlement proxies on average.

Table 1. WTI options with October 2026 expiry: posterior pricing and model-price uncertainty. Pricing error is model price minus settlement price. All price quantities and errors are in U.S. dollars per barrel.

Quantity	N	Value
PI RMSE	310	0.5177
PM plug-in RMSE	310	0.5183
Mean |PI − PM|	310	0.000745
Mean 95% posterior price-interval width	310	0.2733

Source: Authors' calculations from the recorded experiment outputs.

4.3 Historical versus option-implied volatility

Table 2 compares conditional pricing errors for 2,381 contract-date observations across just 15 valuation dates. The historical PI benchmark has MAE 2.6187 and RMSE 3.4090, compared with 0.1075 and 0.1594 for the previous-day APO implied-volatility smile. The expanding earlier-date smile performs similarly. Both specifications have lower errors than historical volatility in all seven expiries. Exploratory bootstrap intervals, calculated by resampling valuation dates, also lie below zero for each expiry (Figure 2).

Table 2. Historical-volatility and earlier option-implied volatility comparisons. Panel A uses 2,381 contract-date observations over 15 valuation dates. Panel B uses only options with an earlier implied-volatility observation for the same contract. Historical PI uses Monte Carlo posterior integration; the other volatility inputs are priced using the same Curran approximation. MAE and RMSE are in U.S. dollars per barrel.

Panel A: Date-ordered conditional pricing comparison

Volatility input	N	Dates	MAE	RMSE
Historical PI	2,381	15	2.6187	3.4090
Rolling 63 returns	2,381	15	4.5201	5.6465
Horizon-matched GARCH(1,1)	2,381	15	2.6730	3.4441
Previous-day APO implied-volatility smile	2,381	15	0.1075	0.1594

Panel B: same-contract lagged-IV benchmark

Volatility input	N	Dates	MAE	RMSE
Historical PI	2,366	15	2.6279	3.4173
Same-contract lagged IV	2,366	15	0.0992	0.1403
Previous-day fitted APO smile	2,366	15	0.1074	0.1594

Source: Authors' calculations from the recorded experiment outputs.

Rolling-window, EWMA, and horizon-matched GARCH(1,1) estimates do not close the gap in pricing errors (Online Supplement Section S1 and Table S1).

Panel B isolates 2,366 observations with an earlier implied volatility for the same contract. Carrying that volatility forward yields lower pooled errors than the fitted previous-day smile (MAE 0.0992 versus 0.1074). On these matched observations, the improvement over historical volatility does not require fitting an implied-volatility smile. The 15 valuation dates are too few to establish that carrying forward implied volatility generally outperforms fitting a smile.

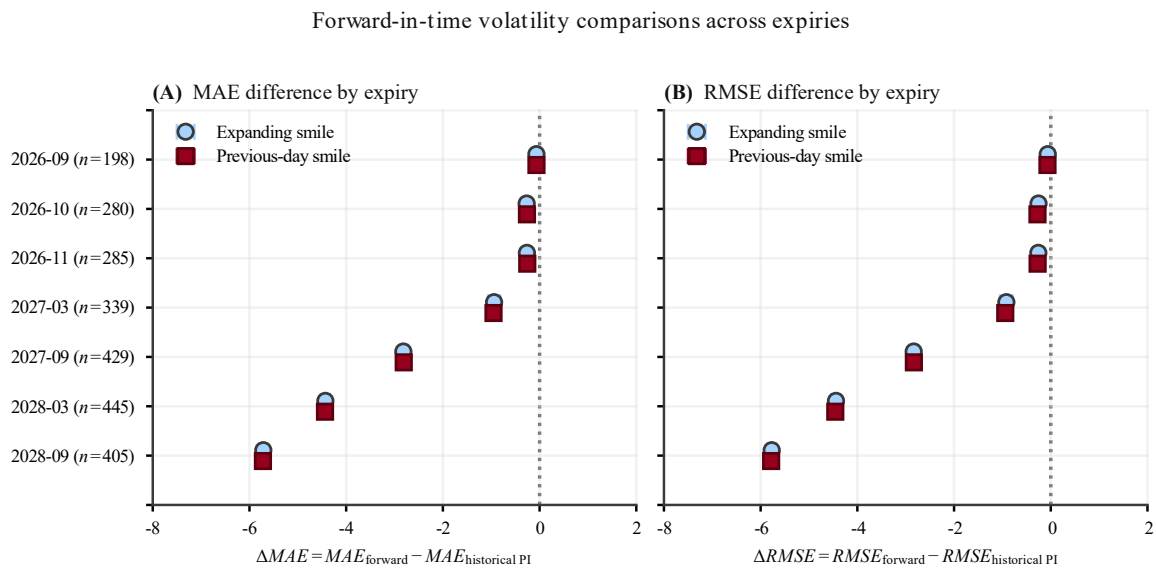


Figure 2. Date-ordered conditional pricing-error differences by APO expiry. Points compare the earlier-date APO implied-volatility specifications with the historical posterior-integrated benchmark; negative values indicate lower error for the implied-volatility specification. Horizontal bars are 95% percentile intervals from valuation-date cluster bootstrap resampling. All implied-volatility inputs are dated before the valuation date in the archived records. We could not verify their original publication times or subsequent revisions.

Source: Authors' calculations from the recorded experiment outputs.

4.4 Cross-instrument comparison using WTI futures options

The cross-instrument comparison uses implied volatility from standard monthly WTI futures options instead of APOs. Table 3 uses 253 October-expiry APO contract-date observations across 10 valuation dates. Historical-volatility posterior integration has MAE 0.4464 and RMSE 0.5562. A previous-day vanilla-WTI implied-volatility specification based on LO options lowers these errors to 0.1559 and 0.2599. The corresponding previous-day APO implied-volatility smile has MAE 0.1676 and RMSE 0.2686.

Table 3. Cross-instrument comparison for 253 October-expiry APO contract-date observations whose moneyness falls within the range covered by earlier LO option data. Historical PI is the canonical Monte Carlo posterior-integrated price; the vanilla-WTI and APO implied-volatility inputs are repriced with the deterministic Curran approximation. All implied-volatility inputs use only earlier-date option information. MAE and RMSE are in U.S. dollars per barrel.

Volatility input	MAE	RMSE
Historical PI	0.4464	0.5562
Previous-day vanilla-WTI IV	0.1559	0.2599
Previous-day APO IV	0.1676	0.2686

Source: Authors' calculations from the recorded experiment outputs.

Valuation-date cluster bootstrap results support the improvement over historical volatility. Differences between LO-implied and APO-implied specifications are small and do not establish consistent superiority by MAE and RMSE. The result shows that earlier implied volatility also produces smaller errors when estimated from a different WTI option product.

4.5 Databento settlement-price results

The separate Databento analysis compares volatility inputs across more valuation dates, but uses a different pricing model from the earlier Curran analysis. Table 4 reports errors for the same observations classified as actual settlements by the data provider. The previous-date AAO implied-volatility benchmark gives an MAE of $0.2184 per barrel, versus $0.8040 for posterior integration of the historical volatility estimate based on earlier changes in the average inferred from put–call parity. Both estimates are evaluated within the same reduced-form terminal-average approximation. Posterior integration itself changes historical proxy pricing little: the PM MAE is $0.8068.

Table 4. Separate Databento AAO pricing comparison across 97,341 observations classified as actual settlements on 189 valuation dates. Errors are in dollars per barrel. The pricing model uses put–call parity estimates from other same-date strikes; it differs from the earlier Curran model based on CL fixing paths.

Volatility input and pricing method	MAE	RMSE
Earlier AAO option-implied volatility	0.2184	0.2979
Historical proxy, posterior integration	0.8040	1.1372
Historical proxy, posterior-mean plug-in	0.8068	1.1401

Source: Authors' calculations from the recorded experiment outputs.

The difference is consistent with the fact that historical changes in the expected average and option-implied volatility of the terminal average measure different quantities. For comparison, we infer implied volatility retrospectively from near-the-money options on the valuation date. Its mean is 30.97% across 1,689 expiry-date groups, compared with 26.68% for the volatility equivalent to the Bayesian plug-in price and 30.88% for earlier AAO implied volatility. Implied volatility inferred from valuation-date prices is used *only to explain* the pricing errors after evaluation, never as an input to the earlier-volatility comparisons. These numbers do not identify a volatility risk premium.

Table 5 asks whether Bayesian modeling of earlier implied volatility improves on simply carrying forward the last observed value, using the same prior option-market information. On 47,718 observations classified as actual settlements across 88 dates in 2025–September 2026, it does not. Carrying forward the last implied volatility gives slightly lower MAE and RMSE. Posterior integration and pricing at the Bayesian posterior mean give nearly identical errors. The estimated improvement in daily MAE from Bayesian updating, averaging equally across valuation dates, is -$0.00100/bbl. A 95% bootstrap interval obtained by resampling calendar months is [-$0.00331,$0.00082]. The interval crosses zero. The observed differences are too small and uncertain to support a claim of economic advantage.

Table 5. Pricing comparison for 47,718 AAO observations classified as actual settlements across 88 valuation dates from 2025 to September 2026. Bayesian estimates use option-implied volatility recorded on earlier dates.

Method	MAE	RMSE
Unchanged earlier-date AAO implied volatility	0.2089	0.3014
Bayesian implied-volatility posterior-mean plug-in	0.2098	0.3023
Bayesian implied-volatility posterior integration	0.2098	0.3024

Source: Authors' calculations from the recorded experiment outputs.

A separate analysis tests the coverage of intervals for settlement-price errors using 89,696 observations classified as actual settlements across 174 dates after an initial estimation period. In the inspected 2025–September 2026 subset, the Bayesian model for pricing errors, combined with the Bayesian implied-volatility price, achieves 86.43% coverage for nominal 90% intervals and 90.11% coverage for nominal 95% intervals. Intervals based on rolling empirical error quantiles, with equal weight across dates, cover 84.72% and 90.66%, respectively. Comparisons of interval scores using calendar-month resampling show no consistent Bayesian advantage. The coverage changes substantially over time: nominal 90% Bayesian intervals cover only 59.0% and 56.9% of targets in March and April 2026, respectively. These are intervals for *errors against settlement prices*, not the posterior ranges of *model prices* reported earlier. The two intervals describe different kinds of uncertainty.

4.6 Robustness, interpretation and limitations

The findings are similar under alternative assumptions about historical returns. A standardized Student-t model leaves the PI–PM difference small relative to historical-volatility pricing errors. Reconstructing first-nearby CL returns contract by contract gives a similar result: on the 2,381 date-ordered observations, historical PI has MAE 2.6468 and RMSE 3.4451, compared with 2.6187 and 3.4090 using the continuous/front-month return proxy. Further historical-volatility benchmarks appear in Online Supplement Section S1.

The small observed PI–PM difference is larger than the numerical approximation error. Monte Carlo with many draws, independently scrambled Sobol simulation, and Curran's approximation agree closely on the price difference. Simulation-based calibration provides a separate check on the posterior implementation. The supplement also examines pricing at the posterior root-mean-square volatility (Section S2), alternative ways to map LO implied volatility to one APO volatility, and the sensitivity to the number of Cox–Ross–Rubinstein tree steps (Section S4). These numerical checks do not establish an exact arbitrage-free representation of the volatility surface.

The market comparisons have important limitations. In the original sample of 2,381 option-date observations, 2,319 have zero recorded trading volume. Earlier option-implied volatility still yields lower errors than the tested historical estimates when we restrict the sample to positive-volume or open-interest observations, but these subsamples are small, especially for some expiries (Online Supplement Section S3). An exchange settlement proxy should not be interpreted as an executable transaction price. Moreover, the main pricing comparison contains only 15 target dates and the external validation only 10. Resampling whole valuation dates accounts for dependence among contracts on the same date, but may not fully account for dependence between nearby dates.

Finally, the one-factor constant-volatility model excludes richer futures-curve dynamics, stochastic volatility, jumps, and convenience yields. The implied volatilities are values inferred under the chosen pricing model. They should not be interpreted as uniquely identified risk-neutral diffusion volatilities or as direct estimates of a variance risk premium. The reported pricing-error differences apply to these volatility inputs and this pricing model; they do not separate physical and risk-neutral sources of volatility structurally.

The Databento analysis covers more dates but has different measurement limitations. Forward averages and effective discount factors are inferred from other *same-date* option strikes; the historical volatility proxy is estimated from earlier parity-implied forward-average changes, not first-nearby CL realized returns; and the lognormal terminal-average approximation replaces the path of known and remaining daily fixings with a model for the terminal average. We cannot independently confirm that the original Barchart end-of-day `Latest` fields are final exchange settlements or executable trade prices. CME Clearing switched AAO end-of-day settlement valuations to its Curran model effective May 18, 2020 (CME Clearing, 2020). Agreement with end-of-day settlement prices may therefore reflect the stability or reproduction of exchange model valuations rather than new information about tradable prices.

A very small price correction need not have economic value. For instance, the older October-expiry mean absolute PI–PM difference is about $0.000745/bbl, well below the standard $0.01/bbl quotation increment. In the longer conditional comparison, Bayesian modeling of implied volatility and simply carrying forward the last observation have very similar errors. Resampling by month does not establish a consistent advantage for either method. The lower-than-intended coverage of pricing-error intervals in early 2026 also cautions against assuming that an uncertainty model fitted to past errors remains well calibrated over time.

The Barchart/LO and Databento analyses differ in the observed prices, data sources, pricing models, available expiries, and information used for valuation. They are *separate analyses*, not pooled observations for one common estimator. Neither tests executable hedging profits or measures a variance risk premium. We still need to verify when the vendors originally published their records, whether those records were revised, whether Barchart data may be redistributed, and whether the original market prices and full input histories can be independently checked. The arithmetic behind the 2,381-observation fitted-smile comparison and the 2,366-observation same-contract comparison has been independently checked, including all 4,762 stored implied-volatility estimates and the recorded ordering of dates. These checks do not establish when the source records first became available.

5. Conclusions and recommendations

We asked whether accounting for uncertainty about volatility changes WTI average-price option values as much as choosing a different volatility input. These are different aspects of the pricing problem. The effect of posterior integration on the point price depends on the uncertainty about volatility and the curvature of the pricing function. The spread of the resulting model prices is a different measure of uncertainty. Our synthetic results illustrate this distinction, while the October-2026-expiry sample shows that posterior integration and pricing at the posterior mean give almost identical point values.

In the original conditional pricing comparison, using earlier option-implied volatility yields smaller errors relative to settlement-price proxies than the tested historical-volatility inputs. For the same contracts, simply carrying forward the last available implied volatility also has lower pooled error than fitting a new smile. The comparison covers only 15 valuation dates and does not establish a general ranking of pricing methods. A separate, longer Databento exercise supports the importance of the volatility input under a different pricing

approximation. In that exercise, Bayesian updating of earlier implied volatility does not reliably improve on simple persistence. Intervals for settlement-price errors based on a Bayesian model also fall below their stated coverage levels in parts of early 2026.

For valuation practice, the main lesson is to compare a posterior-integrated price with a simple plug-in price and to test historical-volatility estimates against available earlier option-implied volatilities. Uncertainty across model prices should be reported separately from intervals for observed pricing errors. The value of more elaborate inference should be judged against the pricing or uncertainty measure it is intended to improve.

These conclusions come from conditional comparisons with end-of-day settlement-price proxies, not from executable quotes or prospective trading results. Some marks may be generated from exchange valuation models; the longer exercise also uses other same-date option prices as inputs. The original study has relatively few valuation dates and many zero-volume observations, and the historical publication times of vendor records remain unverified. Stronger claims about hedging or a volatility risk premium would require reliably time-stamped data, more actively traded contracts, and comparisons with richer futures models.

Data availability

The primary empirical study draws on Barchart WTI Average Price Option end-of-day fields (which can serve as settlement proxies), CL futures contract histories, and Databento standard WTI/LO option data. A distinct longer-date conditional benchmark uses Databento AAO option definitions and settlement statistics; its option forward-average and discount inputs are estimated from other same-date AAO strikes by put–call parity. These inputs are not equivalent to direct contemporaneous CL fixing histories. Source market records require vendor permission before redistribution; first-publication timestamps and historical data revisions are not independently certified. Replication requires independently licensed observations, source-vintage verification and the computational workflow specified in the supporting research records.

Code and results availability

The source-only companion software contains pricing and Bayesian-inference code, synthetic demonstrations, numerical tests, and scripts for the empirical analyses. Its identifying repository location is omitted here to preserve double-blind review and may be provided confidentially to the editor. Re-executing empirical experiments requires separately licensed vendor observations.

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Conflicts of interest

The author declares no conflicts of interest.

References

CME Clearing. (2020). *Valuation model change for four average-price options*. CME Clearing Advisory 20-198, May 15, 2020. https://www.cmegroup.com/notices/clearing/2020/05/Chadv20-198.pdf

CME Group. (2021). *NYMEX Rulebook Chapter 341: WTI Average Price Option*. NYMEX Submission No. 21-201, Exhibit A. https://www.cmegroup.com/market-regulation/rule-filings/2021/6/21-201.pdf

Curran, M. (1994). Valuing asian and portfolio options by conditioning on the geometric mean price. *Management Science*, *40*(12), 1705–1711.

Ewald, C.-O., Wu, Y., & Zhang, A. (2023). Pricing asian options with stochastic convenience yield and jumps. *Quantitative Finance*, *23*(4), 677–692. https://doi.org/10.1080/14697688.2022.2160799

Gilder, D., & Tsiaras, L. (2020). Volatility forecasts embedded in the prices of crude-oil options. *Journal of Futures Markets*, *40*(7), 1127–1159. https://doi.org/10.1002/fut.22114

Guidolin, M., & Timmermann, A. (2003). Option prices under bayesian learning: Implied volatility dynamics and predictive densities. *Journal of Economic Dynamics and Control*, *27*(5), 717–769. https://doi.org/10.1016/S0165-1889(01)00069-0

Kemna, A. G. Z., & Vorst, A. C. F. (1990). A pricing method for options based on average asset values. *Journal of Banking & Finance*, *14*(1), 113–129. https://doi.org/10.1016/0378-4266(90)90039-5

Matías González, A., Martínez-Palacios, M. T. V., & Ortiz-Ramírez, A. (2019). Consumo e inversión óptimos y valuación de opciones asiáticas en un entorno estocástico con

fundamentos microeconómicos y simulación Monte Carlo. *Revista Mexicana de Economía y Finanzas, Nueva Época*, *14*(3), 397–414. https://doi.org/10.21919/remef.v14i3.408

Ortiz Ramírez, A., & Jiménez Preciado, A. L. (2026). Valuación de opciones asiáticas con volatilidad realizada y reversión a la media: Caso CEMEXCPO. *Revista Mexicana de Economía y Finanzas, Nueva Época*, *21*(1). https://doi.org/10.21919/remef.v21i1.1470

Rombouts, J. V. K., & Stentoft, L. (2014). Bayesian option pricing using mixed normal heteroskedasticity models. *Computational Statistics & Data Analysis*, *76*, 588–605. https://doi.org/10.1016/j.csda.2013.06.023

Sagor, H. H., Boone, E. L., & Ghanam, R. A. (2026). Bayesian joint estimation of the hurst parameter and volatility with applications to fractional option pricing. *Risks*, *14*(8). https://doi.org/10.3390/risks14080173

Schwartz, E. S. (1997). The stochastic behavior of commodity prices: Implications for valuation and hedging. *The Journal of Finance*, *52*(3), 923–973. https://doi.org/10.1111/j.1540-6261.1997.tb02721.x

Shiraya, K., & Takahashi, A. (2011). Pricing average options on commodities. *Journal of Futures Markets*, *31*(5), 407–439. https://doi.org/10.1002/fut.20481